

TOMAHAWK AGDM, AB, Canada

WMO: 715660

Lat: 53.4395N

Lon: 114.7189W

Elev: 814

StdP: 91.92

Time Zone: -7.00 (NAM)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-28.9	-25.7	-32.3	0.2	-28.6	-29.1	0.3	-25.3	9.3	0.3	8.2	-2.8	1.9	0	0.634

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.3	27.3	16.9	25.6	16.1	24.0	15.4	18.5	24.9	17.4	23.7	16.4	22.3	3.0	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.2	12.7	21.7	15.0	11.7	20.1	13.9	10.9	18.9	55.9	24.6	52.1	23.4	48.9	22.5	23.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.3	7.1	6.1	DB	-33.0	31.1	3.1	2.3	-35.2	32.7	-37.0	34.1	-38.7	35.4	-40.9	37.1
			WB	-33.1	20.4	2.8	1.4	-35.1	21.4	-36.7	22.3	-38.2	23.1	-40.3	24.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	3.6	-7.9	-8.6	-2.9	3.9	10.6	13.8	16.6	15.1	10.7	4.4	-3.6	-9.5	(6)
(7)		DBStd	10.96	8.14	7.97	6.98	5.00	4.26	2.90	2.94	3.15	4.36	4.45	6.88	7.57	(7)
(8)		HDD10.0	2953	556	521	401	188	44	4	0	2	43	180	408	605	(8)
(9)		HDD18.3	5402	814	754	659	432	242	138	72	107	230	432	658	863	(9)
(10)		CDD10.0	622	0	0	0	7	62	118	203	161	64	6	1	0	(10)
(11)		CDD18.3	28	0	0	0	0	1	3	16	7	1	0	0	0	(11)
(12)		CDH23.3	492	0	0	0	2	42	49	218	125	54	2	0	0	(12)
(13)		CDH26.7	82	0	0	0	0	4	5	43	22	8	0	0	0	(13)
(14)	Wind	WSAvg	2.7	2.5	2.6	2.8	3.2	3.2	2.9	2.6	2.4	2.5	2.7	2.5	2.4	(14)
(15)	Precipitation	PrecAvg	491	25	16	19	31	55	87	94	63	39	24	21	17	(15)
(16)		PrecMax	652	75	30	36	55	110	175	208	138	87	76	61	47	(16)
(17)		PrecMin	321	2	1	3	5	18	25	46	31	9	5	4	5	(17)
(18)		PrecStd	79	17	8	8	15	23	34	39	21	22	17	15	9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.8	9.2	13.6	22.0	27.0	27.4	30.2	29.0	27.5	21.3	14.3	8.1	(19)
(20)		MCWB	3.4	2.9	5.6	10.7	12.9	16.6	18.9	17.6	16.3	11.6	6.9	3.1		(20)
(21)		2%	DB	6.7	5.9	10.5	18.1	24.2	24.7	27.6	26.3	24.8	16.4	9.5	4.8	(21)
(22)		MCWB	2.4	1.4	4.1	8.3	12.2	15.3	17.9	17.0	14.9	9.0	4.4	0.9		(22)
(23)		5%	DB	4.4	3.9	8.0	15.0	21.8	22.7	25.6	24.5	22.0	13.9	6.7	2.6	(23)
(24)		MCWB	0.8	0.1	2.6	6.5	11.2	14.2	17.0	16.6	13.8	7.4	2.8	-0.4		(24)
(25)		10%	DB	2.3	2.0	5.8	12.4	19.2	20.8	23.9	22.6	18.7	11.5	4.4	0.4	(25)
(26)		MCWB	-0.4	-1.1	1.4	5.3	10.0	13.4	16.2	15.7	12.1	6.1	1.2	-1.8		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.2	3.5	6.0	10.7	14.4	17.8	20.7	19.8	17.1	12.1	7.2	3.2	(27)
(28)		MCDWB	8.8	8.3	12.2	21.8	24.1	25.2	26.6	25.5	25.3	20.0	13.1	7.1		(28)
(29)		2%	WB	2.5	1.7	4.3	8.5	12.9	16.3	19.1	18.2	15.4	9.4	4.9	1.4	(29)
(30)		MCDWB	6.3	5.5	10.0	17.1	21.3	22.9	25.5	24.5	23.7	15.3	8.7	4.5		(30)
(31)		5%	WB	1.2	0.4	2.9	7.0	11.8	15.1	17.8	17.1	14.0	8.0	3.2	-0.2	(31)
(32)		MCDWB	4.1	3.5	7.7	14.1	20.0	21.2	24.1	23.0	20.5	12.9	6.3	2.3		(32)
(33)		10%	WB	-0.3	-0.9	1.6	5.7	10.7	14.1	16.7	16.1	12.7	6.6	1.6	-1.8	(33)
(34)		MCDWB	2.3	1.9	5.3	11.5	18.3	19.3	22.6	21.6	18.2	11.0	3.9	0.4		(34)
(35)	Mean Daily Temperature Range		MDBR	9.5	10.2	10.3	10.8	12.3	11.3	12.3	12.2	12.3	10.3	8.2	8.2	(35)
(36)		5% DB	MCDBR	10.8	11.5	13.2	15.7	16.6	14.5	15.1	15.7	18.1	14.8	11.2	9.6	(36)
(37)		MCWBR	7.6	7.8	7.9	8.0	7.4	6.8	7.1	7.6	8.7	8.3	7.2	6.9		(37)
(38)		5% WB	MCDBR	10.8	11.5	12.3	14.4	14.9	12.7	13.9	14.3	16.4	14.1	10.8	9.5	(38)
(39)		MCWBR	7.8	8.0	7.5	7.6	6.9	6.4	7.1	7.5	8.2	8.1	7.2	7.1		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.245	0.258	0.290	0.328	0.343	0.352	0.351	0.357	0.313	0.281	0.254	0.238		(40)
(41)		taud	2.366	2.407	2.420	2.390	2.392	2.382	2.440	2.376	2.500	2.571	2.473	2.333		(41)
(42)		Ebn at Noon	763	870	906	905	905	897	892	864	871	838	753	694		(42)
(43)		Edh at Noon	54	72	89	105	110	113	105	105	82	61	49	46		(43)
(44)	All-Sky Solar Radiation	RadAvg	0.96	1.98	3.44	4.63	5.56	5.92	6.05	5.01	3.48	2.00	1.00	0.69		(44)
(45)		RadStd	0.07	0.15	0.22	0.42	0.40	0.40	0.37	0.34	0.36	0.21	0.09	0.06		(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (51 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page